



Muhammad Afaq Bhatti

AI / ML Engineer

Portfolio User Manual

Complete Guide: Local Setup · GitHub Pages Deployment · Customization

Version 1.0 | 2025

Email: afaqbhatti147@gmail.com

LinkedIn: linkedin.com/in/m-afaq-bhatti

GitHub: github.com/M-Afaq-Bhatti

TABLE OF CONTENTS

What's Inside This Manual

1	Project Overview & File Structure	4
2	Prerequisites & Software Requirements	5
3	Running on Local Machine (localhost)	6
4	Replacing Your Profile Image	8
5	Adding a New Project	9
6	Setting Up the Contact Form	10
7	Deploying to GitHub Pages — Step by Step	11
8	Updating Your Portfolio After Deployment	14
9	Troubleshooting Common Issues	15
10	Customization Reference Guide	16
11	Quick Reference Cheatsheet	17

Note

This manual covers everything you need — from opening the files in VS Code for the first time to having your live portfolio accessible worldwide on GitHub Pages. Follow each chapter in order for the smoothest experience.

CHAPTER 1

Project Overview & File Structure

Understanding what every file does before you touch anything.

What Was Built For You

Your portfolio is a fully static website — meaning it runs directly in the browser with no server, no database, and no backend. It uses HTML5 for structure, Tailwind CSS (loaded from a CDN) for styling, and vanilla JavaScript for interactivity. This makes it extremely fast, free to host, and easy to maintain.

Complete File Structure

```
portfolio/
  index.html ← The entire website lives here
  README.md ← Quick reference guide
  assets/
    images/
      pic2026.jpeg ← YOUR profile photo (replace this)
      quizflow.jpeg ← Project screenshot image
    css/
      style.css ← All custom styles & animations
    js/
      projects.js ← Add new projects here ONLY
      main.js ← All interactivity (do not edit)
```

File Responsibilities

File	Purpose	Edit?
index.html	Main HTML structure — all sections	Rarely
css/style.css	Custom styles, animations, colors	For rebranding
js/projects.js	All project data as a JS array	YES — add projects here
js/main.js	Typing animation, filters, scroll effects	No
assets/images/	Profile photo + project screenshots	YES — add your images

CHAPTER 2

Prerequisites & Software Requirements

Install these tools once — they're free and industry-standard.

Required Software

VS Code	Code editor https://code.visualstudio.com	Free · All platforms
Git	Version control https://git-scm.com	Free · Required for GitHub
GitHub Account	Repository host https://github.com	Free account works perfectly
Live Server (VS Code Extension)	Local preview VS Code Extensions panel	Install inside VS Code

Installing Live Server in VS Code

- 1 Open VS Code**
Launch VS Code on your computer.
- 2 Open Extensions**
Press Ctrl+Shift+X (Windows/Linux) or Cmd+Shift+X (Mac).
- 3 Search**
Type 'Live Server' in the search box.
- 4 Install**
Click Install on the extension by Ritwick Dey (the most popular one).
- 5 Done**
You'll now see a 'Go Live' button in the bottom-right of VS Code.

■ Tip

Git installation also adds 'Git Bash' on Windows — a powerful terminal. You can use it instead of Command Prompt for all commands in this manual.

CHAPTER 3

Running on Local Machine (localhost)

Preview your portfolio live in the browser before deploying.

Method 1 — VS Code + Live Server (Recommended)

This is the best method. It gives you live auto-reload — any file you save instantly updates in the browser with no manual refresh.

- 1 Download the ZIP**
Download your portfolio ZIP from the download link provided. Save it somewhere easy to find, like your Desktop.
- 2 Extract the ZIP**
Right-click the ZIP file → 'Extract All' (Windows) or double-click (Mac). You'll get a folder called 'portfolio'.
- 3 Open in VS Code**
Open VS Code. Go to File → Open Folder → navigate to and select your 'portfolio' folder → click 'Select Folder'.
- 4 Add your images**
Copy your 'pic2026.jpeg' and 'quizflow.jpeg' into the assets/images/ folder. You can drag-and-drop them into VS Code's Explorer panel.
- 5 Launch Live Server**
In VS Code, right-click on 'index.html' in the file explorer on the left → click 'Open with Live Server'.
- 6 View in browser**
Your default browser opens automatically at <http://127.0.0.1:5500>. Your portfolio is running!

■ Tip

Live Server watches ALL files. Edit index.html, style.css, projects.js — save with Ctrl+S — and the browser refreshes instantly. No need to reload manually.

Method 2 — Direct File Open (Quickest)

If you just want to preview quickly without any setup, simply open the 'portfolio' folder in your file manager and double-click index.html. It opens in your default browser. Note: some features like smooth fonts may load slightly differently without a local server.

Method 3 — Python Simple Server (Alternative)

If you have Python installed, open a terminal inside the portfolio folder and run:

```
# Python 3
python -m http.server 8000

# Then open in browser:
http://localhost:8000
```

■ ■ Important

Whichever method you use, the site must always be opened from the 'portfolio' FOLDER — not from inside a sub-folder. Otherwise relative paths (images, CSS, JS) will break.

CHAPTER 4

Replacing Your Profile Image

Make it look like YOU — add your real photo in 60 seconds.

The portfolio is pre-configured to load your profile photo from `assets/images/pic2026.jpeg`. It also has a smart fallback: if the file is missing, it automatically shows a styled 'MA' initials avatar — so the site never breaks.

1 Prepare your photo

Use a clear, professional headshot. Square or portrait orientation works best. JPG or PNG format.

2 Rename the file

Rename your photo file to exactly: `pic2026.jpeg` (lowercase, no spaces). Case matters on GitHub!

3 Copy to images folder

Place the renamed file inside: `portfolio/assets/images/` — overwrite the placeholder if it exists.

4 Verify in browser

Open Live Server and scroll to the Profile section. Your photo appears in the circular frame.

5 Optional — different filename

If you use a different filename, open `index.html` and find the `img` tag in the Profile section. Change `'pic2026.jpeg'` to your filename.

■ Tip

For best results: crop your photo to roughly square (1:1 ratio) before adding it. Recommended size: 400x400px minimum. The CSS automatically applies a circular crop.

Adding Project Screenshot Images

Project images follow the same pattern. Place any project image in `assets/images/` and reference it in `projects.js` like this:

```
image: "assets/images/your-project-screenshot.jpeg"
```

If no image is provided (`image: null`), the project card shows a brain emoji placeholder automatically.

CHAPTER 5

Adding a New Project

The entire projects section is data-driven. Edit ONE file to update everything.

This is the most important workflow to understand. All project data lives in `js/projects.js`. The main website reads this file and dynamically renders every project card, filter button, and category badge. You never edit HTML to add a project.

1 Open projects.js

In VS Code, click on `js/projects.js` in the file explorer.

2 Find the template

Scroll to the bottom — you'll see a commented-out template block starting with `// TEMPLATE`.

3 Copy the template

Copy the entire commented block (from `// {` to `// },`).

4 Paste & uncomment

Paste it after the last project object (before the closing `]`). Remove the `//` comment slashes.

5 Fill in your data

Replace all placeholder values with your real project information.

6 Save the file

Press `Ctrl+S`. Your browser with Live Server instantly shows the new project card.

Project Object Reference

```
{
  id: 2, // Unique number – increment each time
  title: "Your Project Title", // Shown as card heading
  description: "What it does...", // 2-3 sentence description
  techStack: ["Python", "FastAPI"], // Array of technology strings
  category: "AI/ML", // Used for filter buttons
  github: "https://github.com/...", // Your GitHub repo URL
  liveDemo: "https://...", // Live URL or null
  image: "assets/images/img.jpg", // Relative path or null
  featured: false, // true = shows 'Featured' badge
}
```

Available Categories

Full Stack	AI/ML	NLP	Computer Vision	Automation	Other
------------	-------	-----	-----------------	------------	-------

■ Tip

You can create any custom category — just type it in the 'category' field. The filter buttons are generated automatically from whatever categories exist in your data.

CHAPTER 6

Setting Up the Contact Form

Get real emails from recruiters and collaborators in 5 minutes.

The contact form uses Formspree — a free service that forwards form submissions to your email. No backend, no server required. The free plan gives you 50 submissions/month.

1 Sign up for Formspree

Go to <https://formspree.io> and click 'Get Started'. Sign up with your email afaqbhatti147@gmail.com.

2 Create a new form

After login, click '+ New Form'. Give it a name like 'Portfolio Contact'. Click 'Create Form'.

3 Copy your Form ID

Formspree shows you a form ID that looks like: `f/xabc1234`. Copy that entire string including the 'f'.

4 Update index.html

Open `index.html` in VS Code. Press `Ctrl+F` and search for: `YOUR_FORM_ID`. You'll find it in the tag.

5 Paste your Form ID

Replace `YOUR_FORM_ID` with your actual ID. The action attribute should look like:
`action="https://formspree.io/f/xabc1234"`

6 Test it

Save, go to Live Server, fill in the contact form, and submit. Check your email — you should receive the message within seconds!

■ ■ Important

Without setting up Formspree, the form still shows a success message to users (for good UX), but no email is actually sent. Set it up before going live!

CHAPTER 7

Deploying to GitHub Pages

Make your portfolio live on the internet for FREE — permanent URL, HTTPS included.

Phase 1 — Prepare Your Files

- 1 Add your images**
Make sure `pic2026.jpeg` and `quizflow.jpeg` are inside `assets/images/`. These must be present before uploading.
- 2 Set up the contact form**
Complete Chapter 6 to configure Formspree (optional but recommended).
- 3 Test locally**
Run Live Server one final time and check all sections look correct. Verify all links work.

Phase 2 — Create GitHub Repository

- 1 Sign in to GitHub**
Go to <https://github.com> and sign in to your account (or create one — it's free).
- 2 Create new repository**
Click the green 'New' button (top-left) or go to <https://github.com/new>.
- 3 Name your repository**
For a personal portfolio URL (`username.github.io`), name it exactly: `M-Afaq-Bhatti.github.io` Or for a project URL, name it anything like: `portfolio`
- 4 Set visibility**
Select 'Public' — GitHub Pages requires public repositories on free accounts.
- 5 Do NOT initialize**
Leave 'Add README', '.gitignore', and 'license' all UNCHECKED. Click 'Create repository'.

Phase 3 — Upload Your Files Using Git

Open a terminal (Git Bash on Windows, Terminal on Mac/Linux). Navigate to your portfolio folder and run these commands one by one:

```
# Step 1: Navigate to your portfolio folder
cd path/to/your/portfolio

# Step 2: Initialize a git repository
git init

# Step 3: Add all files to staging
```

```
git add .

# Step 4: Create your first commit
git commit -m "Initial portfolio launch"

# Step 5: Rename branch to main
git branch -M main

# Step 6: Connect to your GitHub repository
# Replace M-Afaq-Bhatti with your GitHub username
git remote add origin https://github.com/M-Afaq-Bhatti/M-Afaq-Bhatti.github.io.git

# Step 7: Push files to GitHub
git push -u origin main
```

■ Important

When you run 'git push', GitHub may ask for your credentials. Enter your GitHub username and your Personal Access Token (NOT your password). Generate a token at: GitHub → Settings → Developer Settings → Personal Access Tokens → Tokens (classic).

Phase 4 — Enable GitHub Pages

1 Go to your repository

Open your GitHub repository in the browser: github.com/M-Afaq-Bhatti/M-Afaq-Bhatti.github.io

2 Open Settings

Click the 'Settings' tab at the top of the repository page.

3 Find Pages section

Scroll down in the left sidebar → click 'Pages'.

4 Set source

Under 'Build and deployment' → Source: select 'Deploy from a branch'.

5 Select branch

Branch: select 'main'. Folder: select '/' (root). Click 'Save'.

6 Wait ~2 minutes

GitHub builds your site. A green checkmark appears with your live URL.

7 Your URL is live!

If named M-Afaq-Bhatti.github.io, your portfolio is live at: <https://M-Afaq-Bhatti.github.io>

■ Tip

First deployment can take 2-5 minutes. Refresh the Pages settings tab until you see: 'Your site is live at https://...' with a green checkmark.

CHAPTER 8

Updating Your Portfolio After Deployment

Push changes live in under 2 minutes.

Every time you add a new project, update your bio, or change any file, use these three Git commands to push the update live:

```
# From inside your portfolio folder:

# Stage all changed files
git add .

# Commit with a descriptive message
git commit -m "Add new project: Your Project Name"

# Push to GitHub (site updates in ~60 seconds)
git push
```

Typical Update Workflow

- 1 Make your changes**
Edit projects.js to add a project, or any other file.
- 2 Test locally**
Check with Live Server that everything looks correct.
- 3 Open terminal**
Navigate to your portfolio folder in terminal.
- 4 Run the 3 commands**
git add . → git commit -m "message" → git push
- 5 Wait 60 seconds**
GitHub Pages rebuilds automatically. Refresh your live URL.

■ ■ Note

You can also upload files directly through the GitHub website (drag and drop) without using terminal commands. But using Git is the professional way and gives you full version history.

CHAPTER 9

Troubleshooting Common Issues

Solutions to the most frequent problems.

■■ Important

Images not showing: Check the filename is exactly correct (case-sensitive). 'Pic2026.JPEG' ≠ 'pic2026.jpeg'. Also confirm the file is in assets/images/ — not in the root portfolio folder.

■■ Important

CSS/styles not loading: Make sure you opened the FOLDER in VS Code, not just the index.html file directly. The relative path css/style.css needs the portfolio folder as the root.

■■ Important

GitHub Pages shows 404 error: The repository must be Public, not Private. Also verify your source branch is 'main' and folder is '/' (root) in GitHub Pages settings.

■■ Important

Contact form not sending email: Check that you replaced YOUR_FORM_ID with your real Formspree ID. Also verify you confirmed your email address in Formspree after signing up.

■■ Note

Typing animation not working: This requires JavaScript to be enabled in your browser. All modern browsers support this by default. Try a hard refresh: Ctrl+Shift+R (Windows) or Cmd+Shift+R (Mac).

■■ Note

Git push asks for password: GitHub no longer accepts account passwords for Git operations. You need a Personal Access Token. Generate one at: GitHub Settings → Developer Settings → Personal Access Tokens.

■■ Note

Changes not showing after git push: GitHub Pages can take 1-3 minutes to rebuild. Also try clearing browser cache: Ctrl+Shift+Delete and clear cached images and files.

■ Tip

Project card image is broken: No problem — if an image fails to load, the card automatically shows a brain emoji placeholder. Add the correct image path in `projects.js` when ready.

CHAPTER 10

Customization Reference Guide

Make the portfolio truly yours.

Changing the Accent Color

Open `css/style.css`. At the top you'll find the CSS variables block. Change `--accent` to any hex color:

```
:root {  
  --accent: #2563eb; /* Blue (default) */  
  --accent-light: #3b82f6; /* Lighter variant */  
  --accent-dark: #1d4ed8; /* Darker variant */  
  --accent-muted: #eff6ff; /* Background tint */  
}  
  
/* Popular alternatives: */  
/* Purple: #7c3aed / #8b5cf6 / #ede9fe */  
/* Teal: #0d9488 / #14b8a6 / #f0fdfa */  
/* Emerald: #059669 / #10b981 / #ecfdf5 */
```

Changing Fonts

The portfolio uses Syne (headings) + DM Sans (body) from Google Fonts. To change them, update the `@import` line in `style.css` and the font-family variables:

```
/* In css/style.css – top of file */  
  
@import url('https://fonts.googleapis.com/css2?family=YOUR_FONT:wght@400;700&display=swap');  
  
:root {  
  --font-display: 'Your Font', sans-serif;  
  --font-body: 'Your Body Font', sans-serif;  
}
```

Updating Personal Info

What to change	Where to find it	Search for
Name, role, email	<code>index.html</code>	Muhammad Afaq Bhatti

Tagline	js/main.js	typingTexts = [
About text	index.html	about-quote
Skill names & percentages	index.html	skill-name
Social links	index.html	github.com / linkedin.com
All projects	js/projects.js	const projects = [

CHAPTER 11

Quick Reference Cheatsheet

Everything you need on one page.

Essential Git Commands

<code>git init</code>	Initialize a new Git repository in current folder
<code>git add .</code>	Stage ALL changed files for commit
<code>git commit -m "msg"</code>	Save staged changes with a description
<code>git push</code>	Upload commits to GitHub
<code>git status</code>	See which files have been changed
<code>git log --oneline</code>	Show commit history
<code>git remote add origin URL</code>	Connect local repo to GitHub repo

Key URLs

Resource	URL
Your GitHub Profile	https://github.com/M-Afaq-Bhatti
Your LinkedIn	https://www.linkedin.com/in/m-afaq-bhatti/
Contact Email	afaqbhatti147@gmail.com
Formspree (contact form)	https://formspree.io
GitHub Pages docs	https://pages.github.com
Tailwind CSS docs	https://tailwindcss.com/docs
Google Fonts	https://fonts.google.com

■ Tip

Congratulations on your new portfolio! You now have a premium, production-ready website that showcases your AI/ML expertise. Keep adding projects as you build them — each one makes your profile stronger. Good luck with your applications! ■